

```
a = int(input("Enter first number: "))
Enter first number: 5
b = int(input("Enter second number: "))
Enter second number: 6

if b > a:
    print("b is greater than a")
elif b < a:
    print("b is less than a")
else:
    print("b is equals to a")

b is greater than a
,
```

```
a = int(input("Enter first number: "))
Enter first number: 8
b = int(input("Enter second number: "))
Enter second number: 6
c = int(input("Enter third number: "))
Enter third number: 4

if a>b and b>c:
    print("both conditions are true")
else:
    print("conditions are false")

both conditions are true
.
```

Decision Making Structure Tasks:

- ❑ Write program to check if the given number is positive, negative or zero.
- ❑ Ask the user for a grade percentage and display the corresponding letter grade (A, B, C, D, F)

A-Grade Range (80(inclusive) - 100(inclusive))

B-Grade Range (70(inclusive) - 80(exclusive))

C-Grade Range (60(inclusive) - 70(exclusive))

D-Grade Range (50(inclusive) - 60(exclusive))

Below 50 Fail

- ❑ Write program that displays: Kamran Akmal on output, if score > 30, Shoaib Akhtar, if $20 < \text{score} < 30$ and Shahid Afridi, if $10 < \text{score} < 20$.

Decision Making Structure Task:

- ❑ Write a program that takes positive integer as input from user and checks whether the number is even or odd and display the appropriate message on screen.

- Iterative Structure (For, While Loops)

- A loop is a programming structure that repeats a block of code multiple times.
- For loop is used to iterate over a sequence (that is either a list, a tuple, a dictionary, a set, or a string) – how many times you want to repeat the block of code.
- While loop is used to repeat block of code until condition becomes False.

For loop structure:

for variable_name in sequence:
 # block of code

While loop structure

while condition:
 # block of code

- **For loop Tasks:**

1. Write a program to print even numbers from 2 to 20.

(Hint: Use range function in for loop)

2. Write a program to print odd numbers from 0 to 20.

(Hint: Use range function in for loop)

3. Print multiplication table of the number input by user.

4. Compute the sum of first 10 natural number.

(Hint: you can use range function to generate the sequence of natural numbers and then iterate for loop on that sequence)

- **While loop Tasks:**

1. Write a program to print numbers in descending order from 10 to 1.
(Hint: Use range function in for loop)
2. Write a program to keep asking the user to input numbers and add them until they enter 0.
(Hint:
 - a. you must create a variable before while loop in which you will have to keep adding the input number using arithmetic assignment operator.*
 - b. you have to use if keyword inside the body of while loop to check if the input number is 0)*
3. Write a program to keep asking the user to enter correct password (e.g. "python123") until they entered it right.
4. Write a program to keep asking the user to guess a number until they guess correct number (i.e. 7).
(Hint: same as hint (b) in question 2)